

# Recommended implementation plan

## 1) Set up a single email service layer (backend)

- Create `functions/src/email/brevoClient.ts` with:
    - Brevo API key from Firebase Functions config / Secret Manager (never hardcoded)
    - `sendEmail({to, templateId, params, tags})`
    - `sendBulkEmail({recipients, templateId, paramsPerUser})`
  - Add retry + error logging wrapper.
  - Store every send attempt in Firestore ( `email_activity` ) with status, template, recipient, error.
- 

## 2) Define email templates in Brevo first

Create reusable templates for:

- `course_inactive_7_days`
- `course_inactive_14_days`
- `graduation_meeting_time_selected`
- `graduation_application_accepted`
- `graduation_application_rejected`
- `event_announcement_to_registrants`
- `admin_custom_announcement`
- `app_access_code_invite`

Each template should use vars

like `{{first_name}}` , `{{meeting_time}}` , `{{event_title}}` , `{{invite_code}}` , etc.

---

### 3) Add an email preference model (required)

In `users/{uid}` add:

- `email_opt_out_all: boolean`
- optional granular prefs:
  - `email_pref_course_nudges`
  - `email_pref_graduation_updates`
  - `email_pref_events`
  - `email_pref_admin_messages`

Check these before every send.

---

### 4) Implement each trigger/use case

#### A) Incomplete course nudges (1 week + 2 weeks)

- Scheduled function daily (or every 12h):
  - Query users with active enrollment
  - Calculate “last progress timestamp” from your progress source (`courseProgress`, `user_progress`, lesson events)
  - If no progress for 7+ days and 14+ days, send corresponding template
- Add idempotency markers to avoid repeat spam:
  - `email_nudges.course_inactive_7d_sent_at`
  - `email_nudges.course_inactive_14d_sent_at`

#### B) Graduation application updates

Trigger on `GraduationApplications/{id}` updates:

- When admin sets selected time → send `graduation_meeting_time_selected`
- When status becomes accepted/rejected → send corresponding template
- Also when “admit to alumni” action completes (if separate from accepted), send separate email.

- Use transition checks ( `before` vs `after` ) so you only send once per state change.

## C) Batch emails to event registrants

- Add callable `adminSendEventRegistrantEmail`
  - Input: `eventId` , subject/template, content params
  - Load `registered_users` from `events` / `events_mobile`
  - Resolve to user emails
  - Send via Brevo bulk API in chunks
- Log campaign in `email_campaigns` + per-recipient results in `email_activity` .

## D) Admin ability to send emails

- Add callable `adminSendEmail`
  - Audience options:
    - specific emails
    - specific UIDs
    - users by role
    - all users (guarded)
  - Validate admin role ( `Admin` / `superAdmin` )
  - Use approved template IDs or safe HTML/plain body policy
- Add an Admin panel page with:
  - template picker
  - audience builder
  - preview
  - dry-run count
  - send + status history

## E) Send app access code emails

- In your existing invite/code flow ( `createOrUpdateEligibleUser` , `generateInviteCode` , etc.):
    - After code generated, trigger/send `app_access_code_invite`
    - Include code + redemption steps + expiration/revocation info
  - Also expose a “resend invite email” admin action.
- 

## 5) Security + reliability hardening

- Use Firebase Secret Manager for Brevo key.
  - Rate-limit admin send callables.
  - Chunk batch sends (e.g., 100–500 recipients/request depending on Brevo limits).
  - Add dead-letter retry mechanism for transient API failures.
  - Add structured logs + metrics dashboard:
    - sent, failed, bounced, deferred, open/click (if webhook ingestion enabled).
- 

## 6) Brevo webhook ingestion (strongly recommended)

Create HTTPS endpoint/function to receive Brevo events:

- delivered
  - open
  - click
  - bounce
  - blocked Persist into Firestore ( `email_delivery_events` ) and tie back to campaign/activity IDs.
- 

## 7) Rollout strategy

1. Implement service layer + one template + one trigger first (graduation updates easiest).
2. Validate in stage with test users.

3. Add scheduled inactivity nudges.
4. Add admin send + event registrant batch send.
5. Add webhook tracking and reporting.